Machine Learning Prediction of Structural Load Limits in Damaged Wings

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1. Title and Team

1.1. Project title

This project is the “Machine Learning Prediction of Structural Load Limits in Damaged Wings” for the Spring 2026 AERO 489 “Special Topics: Applied Machine Learning” course.

1.2. Team members and roles

Our team includes:

- Barrett Brown: Scripting & Simulation (Data Production)
- Kirin Chadha: Classical Machine Learning Approaches
- Malachi Drew: Aircraft Design and Modeling & Classical Machine Learning Approaches
- Ian Wilhite: Modern Machine Learning Approaches
- Adam Zheng: FEA Modeling and Meshing

2. Problem Statement

2.1. What aerospace problem are you solving?

As aircraft continue to see deployment in contested environments, they are likely to receive damage, and pilots must perform complex evasive maneuvers to prevent continued damage and continue operation. Pilots must assess the damage to their aircraft accurately through Structural Health Monitoring (SHM) in order to effectively plan these maneuvers and prevent critical aircraft failure while utilizing their full flight capabilities, requiring a quantitative estimation of maneuverable capabilities using measurable data onboard the aircraft.

2.2. Why is it important for aerospace engineering or operations?

Aircraft continue to be utilized in contested environments in which damage can often be sustained to an aircraft, causing critical mission failure, risking high-value assets, and causing undue loss of life. In the short term, there is a need for human pilots to assess their damage and capable maneuverability, in the long term, there is a need to design data-informed SHM systems that can be integrated into future autonomous intelligent aircraft, enabling them to revolutionize deployment in contested domains.

2.3. What are the engineering consequences if the problem is not solved?

Aircraft operating in these contested environments without proper ability to estimate their sustained damage and maximum viable maneuverability remain vulnerable to overestimation of modified capabilities resulting in critical failures and underestimation of modified capabilities resulting in overly-conservative trajectory planning in time-critical deployments.

3. Background and Motivation

3.1. Brief context and relevant existing work.

Current state of the art Structural Health Monitoring (SHM) in aerospace has largely been guided by targeted preventative maintenance, failure detection of sensors, and the automation of inspection tasks of critical infrastructure [1]. These methods increasingly utilize machine learning for interpretation, and current work includes leveraging model based methods to improve physical interpretability, extrapolation, and decrease training time [2]. Relevant to this project, a group at NUST previously utilized in flight strain data to predict fatigue damage within the aircraft structure [3].

3.2. Why is the problem challenging today?

In-flight damage assessment and load prediction requires estimating many complex internal and external dynamic factors, which requires gathering datasets of many modified aircraft that are subjected to severe flight-altering damage, then experimentally determining their new maximum g-limit. These costs lead to a large issue of data scarcity, indicating utility in simulated data and physically informed methods that perform better on smaller datasets.

3.3. What gap does this project address?

Existing works focus on long-term aircraft maintenance, not real-time damage estimation. The proposed problem addresses the need for in-flight understanding of major flight-altering damage, and leveraging simulation to perform this estimation with limited data.

4. Proposed Approach

4.1. What machine learning or hybrid modeling method will you use?

Derived Features:

A. Tip Deflection/Acceleration [m / m/s^2]
B. Average Strain Change Factor [mm/mm]
C. Strain energy $\frac{1}{2} \cdot \Sigma(\varepsilon_i^2 \cdot E \cdot V_i)$ [J]

Classical ML Approaches:

1) Linear regression on derived features
2) Polynomial regression on derived features
3) Gaussian Process Regression (GPR)
4) Random Forest (RF)

Modern ML Approaches:

5) Feedforward Neural Network (NN)
6) Physically Informed Neural Network (PINN)
7) Deep Learning

4.2. What are the model inputs and outputs?

- Time series Inputs:
    - 20 strain gauges per wing
    - Wing tip deflection
    - Total aircraft acceleration
- Constant Inputs:
    - Locations of strain gauges (xyz wrt. airframe)
    - Full aircraft acc (gs) -> maybe constant maybe variable
        - 2*F = mg*k, k is the number of g's the aircraft pulls, F is the force on one wing. Force is in the direction of lift.
- Output
    - Max allowable G loading before failure

4.3. Why is this approach appropriate for the problem?

We choose to leverage simulated data due to the data scarcity in datasets of aircraft damage, and the fidelity of modern simulation tools. The usage of strain gauges mirrors the cost effectiveness of the tooling required, and the amount of useful data able to be gathered. The shear loading across an aircraft wing is well studied (Fig. 1), and therefore provides analytical governing equations and models to seed physically derived Machine Learning (ML) methods. ML methods should therefore be able to identify the trends from the loading conditions applied, and successfully approximate the loading condition before failure.

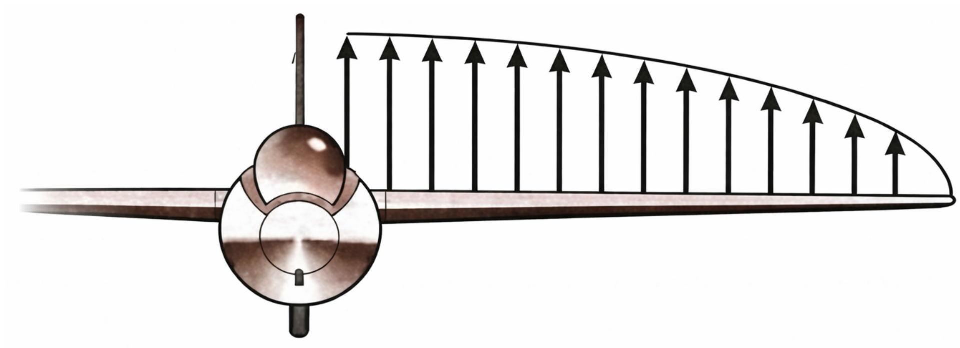

Figure 1: Shear forces acting on the wing.

5. Data and Methods

5.1. What data sources will you use?

We plan to use data from a purpose built simulation in ABAQUS that will give us strain and deflection information for the damaged aircraft wing. These simulations will be conducted by intersecting 20 bullet trajectories with the internal structure of the wing and removing the associated material, as shown in Figure 2. While this is still an approximation (a bullet would cause more damage than a clean removal of material), this method will quickly create training data for the ML algorithms that is close to an approximation of real life. Unfortunately, we will not be able to include the effect of the bullet holes on the skin and the interaction between the skin and the internal structure of the wing. This approach relies on the internal structure of the model, and neglects the structural integrity contributions from the skin.

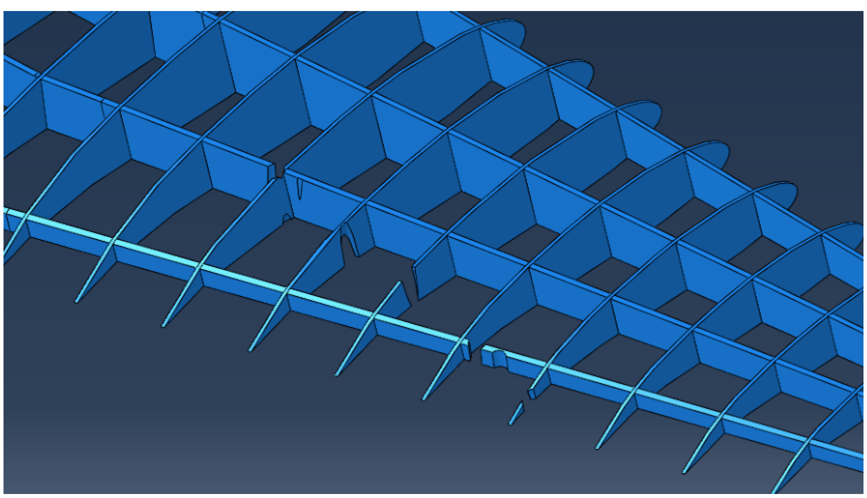

Figure 2: Example of a damaged wing

5.2. How will you collect, preprocess, and validate the data?

We will collect our data from the ABAQUS simulation. Our outputs from the ABAQUS simulations will include the wing tip deflection at failure [1], wing load at failure, and the location and measurements of 20 strain gauges throughout the wing at a range of loads. This "time series" data will give us the ability to predict the maximum load even when the wing does not experience these loads, as in a real-world situation you cannot find the maximum load experimentally. So, for each simulation, the useful output data we will get is:

- Tip Deflection
  - ~20 load points between unloaded and failure
- 20 strain gauges
  - ~ 20 load points between unloaded and failure

This data cannot be validated, as we do not have access to an A-10 Warthog or its wing, nor do we have the ability to shoot it with an anti-aircraft gun; however, we believe that this will be an adequate approximation of what this data would look like in real life.

5.3. What performance metrics will you use to evaluate success?

We will use adjusted R^2, MAE, and RMSE to determine how well the models predict in the test data. We will also compare performance in the training data to catch any overfitting that may occur. Since this is a safety critical system, we will report the maximum overprediction made by each model (predicted g-limit - FEA modeled g-limit). We will also use required margin of safety (MOS) as a performance metric, where the required MOS for a model is such that under 1% of predictions will exceed the actual g-limit after subtracting the MOS from the predicted g-limit.

Another important factor is the inference time for each model, since this should be computed real-time on the aircraft. Since many details may differ between this model and a deployed model, we will use inference times on the same computer hardware to compare each model to each other, but will not compare to external benchmarks or requirements.

6. Expected Outcomes

6.1. What do you expect to deliver at the end of the project?

We expect to deliver several machine learning models capable of predicting max g-loads for a damaged aircraft wing. Models will use a variety of classical and modern machine learning techniques, and will require minimal inputs as could be measured on an actual combat aircraft (with minimal added complexity/weight). Analysis of each model's efficacy will be presented based on its performance on test data (developed in the same way as the training data).

[1] We define failure as when any point of the wing reaches the elastic limit of the material.

6.2. How will you know if the project succeeded?

In short, project success will be defined by the development of one or more models that reliably predict the modified g-limit for a damaged wing. The following requirements must be met for a model to qualify as accurate and reliable:

- Adjusted $R^2 > 0.8$
- RMSE < 0.75g
- MAE < 0.5g
- Required MOS < 0.25g
- Similar performance on test & training sets

7. Timeline and Milestones

Our timeline is as follows:

- [x] ~~4/19: Working Abaqus sim due~~
- [x] ~~4/19: Proposal Submission~~
- [x] ~~4/21: Abaqus Data Processing Due~~
- [x] ~~4/22: Abaqus Runs on HRBB Computers (WIP)~~
- [ ] 4/24: Preliminary ML scripts due (without data) (parallelized)
- [ ] 4/27: ML Analysis & Training Due
- [ ] 4/29: **Final Check-in** & Report Finalization
- [ ] 5/1: Project Submission

8. Risks and Mitigation

8.1. What are the main risks or challenges?

The biggest risks in this project relate to insufficiency of data. In the deployment of this model, data would be much more limited than in our FEA simulations (20 strain gauges instead of a whole stress/strain field). There is a risk that the data provided will be insufficient to accurately predict the load limit of the structure across varied damage types and positions.

The second major risk is that our simulations will be too computational intensive to produce sufficient training data. Since we're developing training data from conventional simulations instead of just pulling from the real world, each datapoint will require significant computational resources. If the training data requirement is too large, we will be unable to meet it.

8.2. How will you mitigate them?

To ensure the obtainable data at deployment is sufficient, we'll implement a two-pronged approach. First, we will match the training data to the expected real world data. Instead of inputting a large amount of stress and strain data, we will process the simulation results to pull out only strain at 20 points, tip deflection, and g-limit (derived from load at failure). Second, we will disperse the strain gauges across the whole wing in all three dimensions. This will ensure data collected has relevance to the whole structure, despite its limits.

To minimize the risk in developing training data, we have opted for a skinless structure and are using FEA best practices to reduce computational requirements (localized mesh refinement around holes, etc). We have also obtained access to the High Performance Research Computer (HPRC) and automated most of the simulation process to decrease computation time and increase the number of simulations we can run (more training data).

9. Aerospace Impact

9.1. Why does this matter for aerospace engineering specifically?

In aerospace environments, real-time assessment of structural limits is uniquely valuable. When flying an aircraft, pausing for lengthy analysis isn't possible: the plane has to keep flying, and decisions have to be made in moments. Pilots with real-time structural health information (such as a real-time g-limit) will be better equipped to fly in demanding off-nominal conditions while maintaining both safety and maximum performance.

9.2. How will the result translate to aerospace applications?

If successful in developing a reliable model for g-limit prediction, this project will serve as the first step toward an integrated SHM system in combat aircraft. By installing minimal sensors (small strain-gauges) and leveraging existing information (g-meter), aircraft designers can collect all the information needed to deploy such a model on new or existing aircraft. While a new model will be required for each aircraft, this should be trivial given existing structural analysis requirements. Once deployed, any pilot (or control scheme) in a damaged aircraft could quickly reference the updated g-limit and ensure its maneuvers remained within the modified limit while exiting the fight and returning to base.

References

[1] Machine Learning for Structural Health Monitoring of Aerospace Structures: A Review https://www.mdpi.com/1424-8220/25/19/6136

[2] Physics-informed machine learning for Structural Health Monitoring https://arxiv.org/pdf/2206.15303

[3] Integrated engineering framework for fatigue damage prediction of fighter aircraft using machine learning https://www.sciencedirect.com/science/article/pii/S2590123025037661